



CAT 2025 Slot 1 Question Paper - QA

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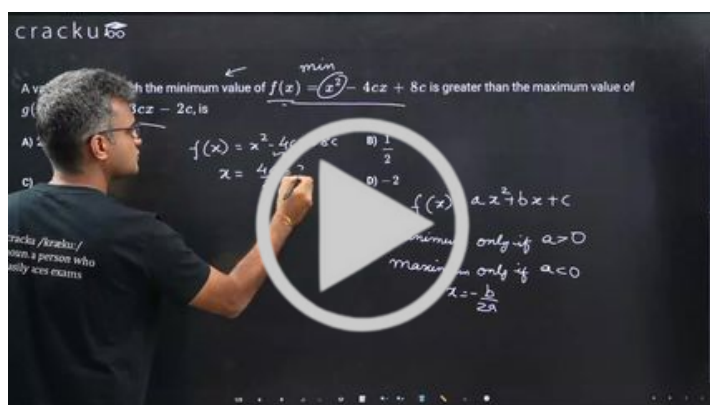
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Quant

1. A value of c for which the minimum value of $f(x) = x^2 - 4cx + 8c$ is greater than the maximum value of $g(x) = -x^2 + 3cx - 2c$, is

- A 2
- B $\frac{1}{2}$
- C $-\frac{1}{2}$
- D -2



VIDEO SOLUTION

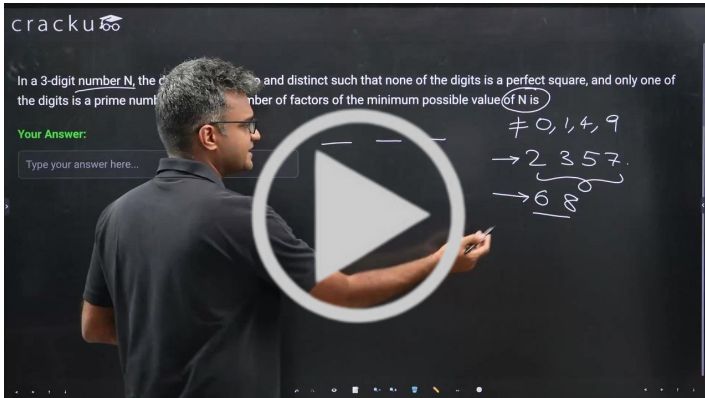
2. Shruti travels a distance of 224 km in four parts for a total travel time of 3 hours. Her speeds in these four parts follow an arithmetic progression, and the corresponding time taken to cover these four parts follow another arithmetic progression. If she travels at a speed of 960 meters per minute for 30 minutes to cover the first part, then the distance, in meters, she travels in the fourth part is

- A 76800
- B 112000
- C 96000
- D 86400



VIDEO SOLUTION

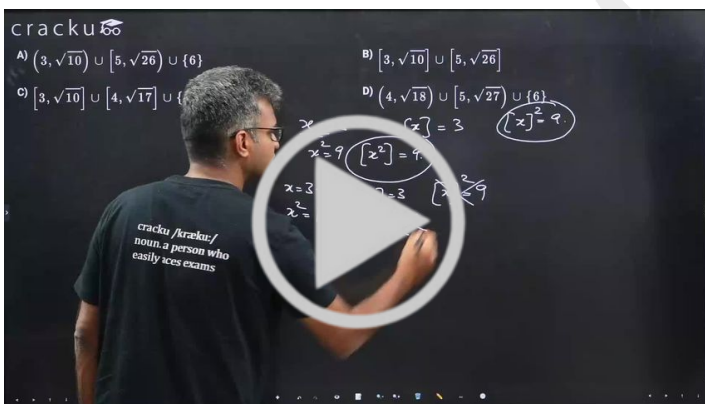
3. In a 3-digit number N , the digits are non-zero and distinct such that none of the digits is a perfect square, and only one of the digits is a prime number. Then, the number of factors of the minimum possible value of N is



VIDEO SOLUTION

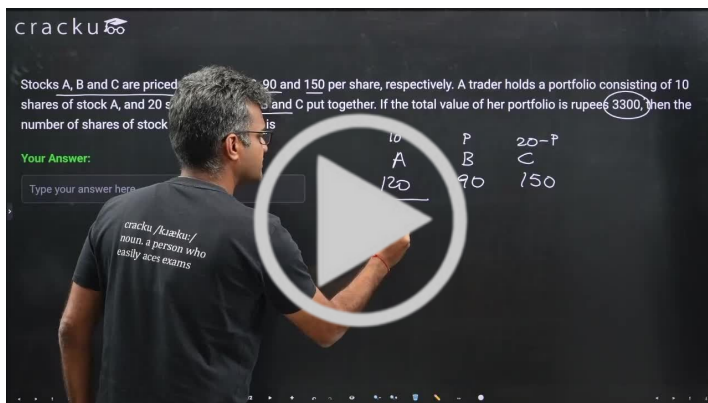
4. Let $3 \leq x \leq 6$ and $[x^2] = [x]^2$, where $[x]$ is the greatest integer not exceeding x . If set S represents all feasible values of x , then a possible subset of S is

- A $(3, \sqrt{10}) \cup [5, \sqrt{26}) \cup \{6\}$
B $[3, \sqrt{10}] \cup [5, \sqrt{26}]$
C $[3, \sqrt{10}] \cup [4, \sqrt{17}] \cup \{6\}$
D $(4, \sqrt{18}) \cup [5, \sqrt{27}) \cup \{6\}$



VIDEO SOLUTION

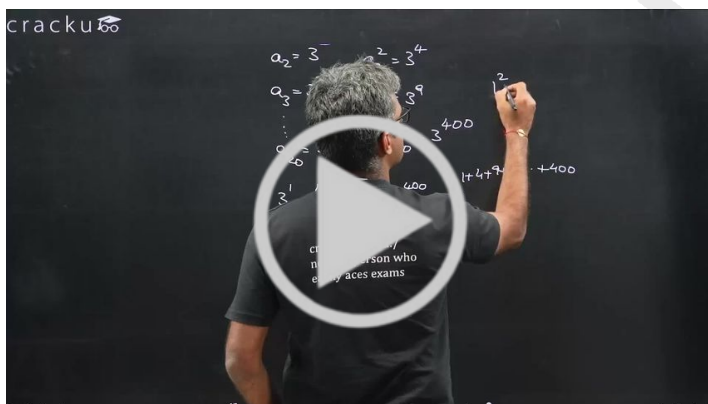
5. Stocks A, B and C are priced at rupees 120, 90 and 150 per share, respectively. A trader holds a portfolio consisting of 10 shares of stock A, and 20 shares of stocks B and C put together. If the total value of her portfolio is rupees 3300, then the number of shares of stock B that she holds, is



VIDEO SOLUTION

6. For any natural number k , let $a_k = 3^k$. The smallest natural number m for which $\{(a_1)^1 \times (a_2)^2 \times \dots \times (a_{20})^{20}\} < \{a_{21} \times a_{22} \times \dots \times a_{20+m}\}$, is

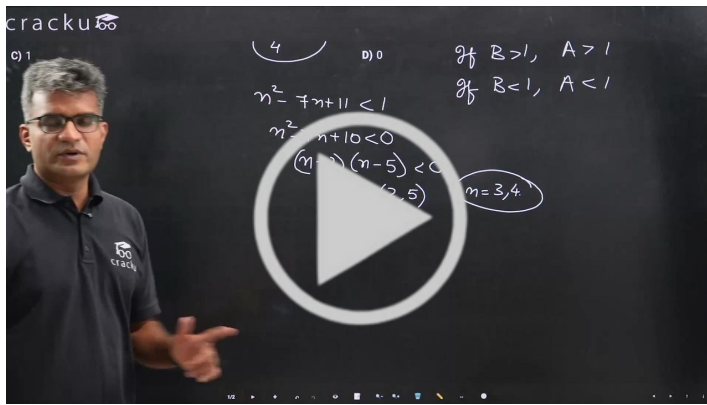
- A 58
- B 59
- C 56
- D 57



VIDEO SOLUTION

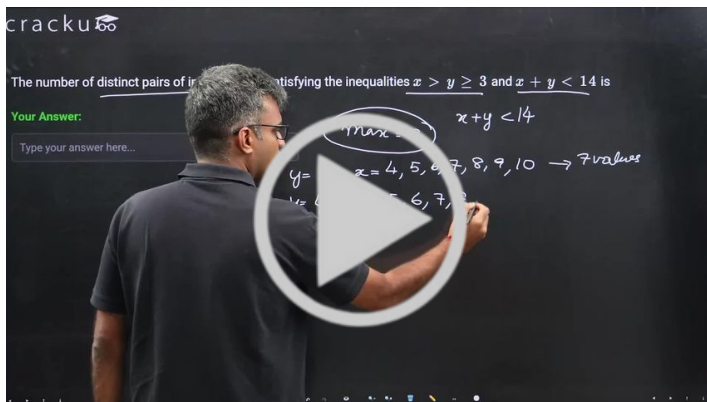
7. The number of distinct integers n for which $\log_{\frac{1}{4}}(n^2 - 7n + 11) > 0$, is

- A 2
- B infinite
- C 1
- D 0



VIDEO SOLUTION

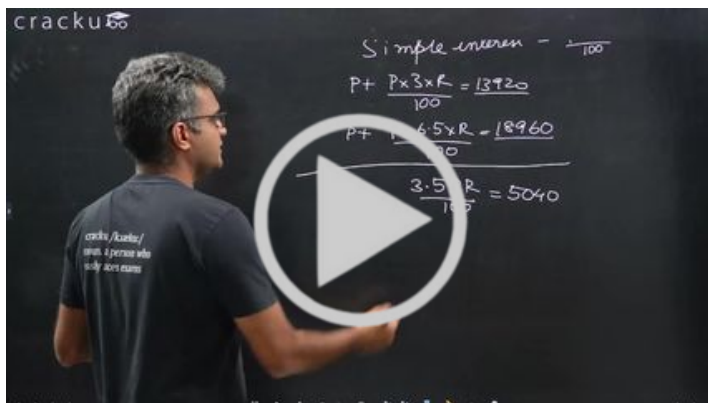
8. The number of distinct pairs of integers (x, y) satisfying the inequalities $x > y \geq 3$ and $x + y < 14$ is



VIDEO SOLUTION

9. At a certain simple rate of interest, a given sum amounts to Rs 13920 in 3 years, and to Rs 18960 in 6 years and 6 months. If the same given sum had been invested for 2 years at the same rate as before but with interest compounded every 6 months, then the total interest earned, in rupees, would have been nearest to

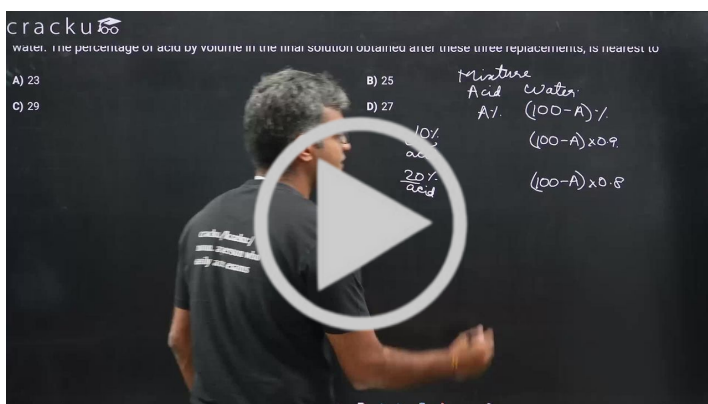
- A 3221
- B 3180
- C 3150
- D 3096



VIDEO SOLUTION

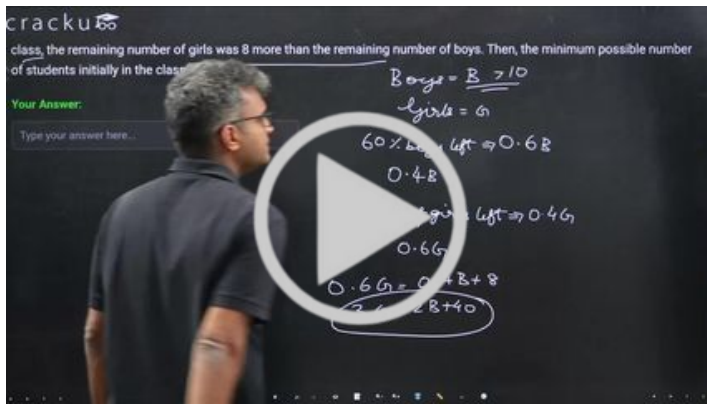
10. A container holds 200 litres of a solution of acid and water, having 30% acid by volume. Atul replaces 20% of this solution with water, then replaces 10% of the resulting solution with acid, and finally replaces 15% of the solution thus obtained, with water. The percentage of acid by volume in the final solution obtained after these three replacements, is nearest to

- A 23
- B 25
- C 29
- D 27



VIDEO SOLUTION

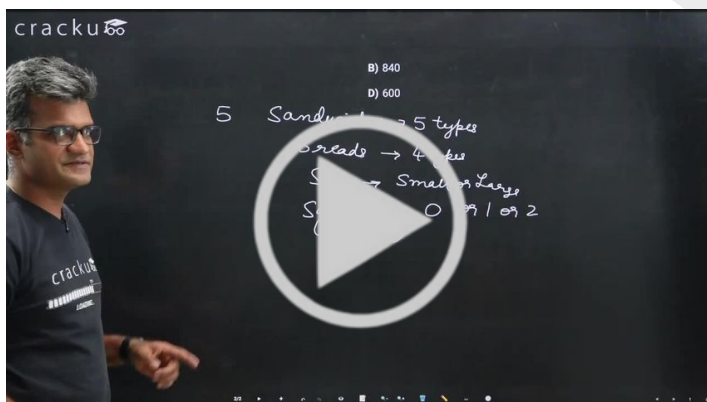
11. In a class, there were more than 10 boys and a certain number of girls. After 40% of the girls and 60% of the boys left the class, the remaining number of girls was 8 more than the remaining number of boys. Then, the minimum possible number of students initially in the class was



VIDEO SOLUTION

12. A cafeteria offers 5 types of sandwiches. Moreover, for each type of sandwich, a customer can choose one of 4 breads and opt for either small or large sized sandwich. Optionally, the customer may also add up to 2 out of 6 available sauces. The number of different ways in which an order can be placed for a sandwich, is

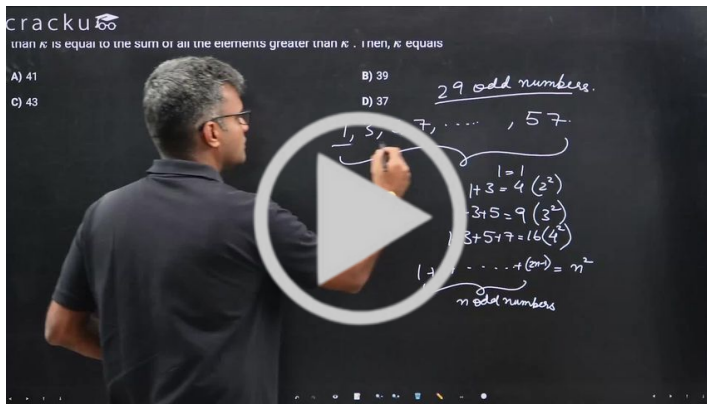
- A 880
- B 840
- C 800
- D 600



VIDEO SOLUTION

13. In the set of consecutive odd numbers $\{1, 3, 5, \dots, 57\}$, there is a number k such that the sum of all the elements less than k is equal to the sum of all the elements greater than k . Then, k equals

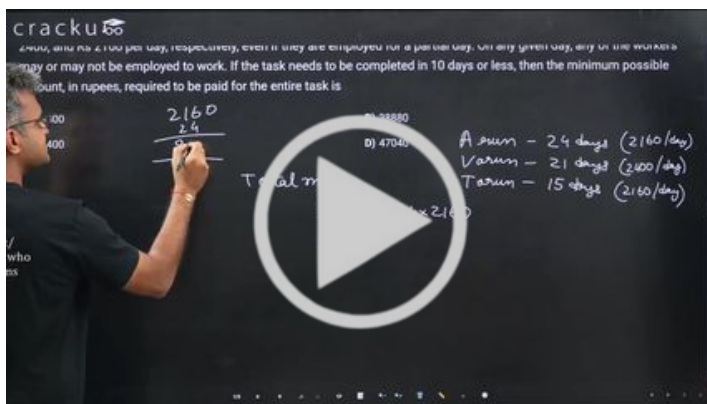
- A 41
- B 39
- C 43
- D 37



VIDEO SOLUTION

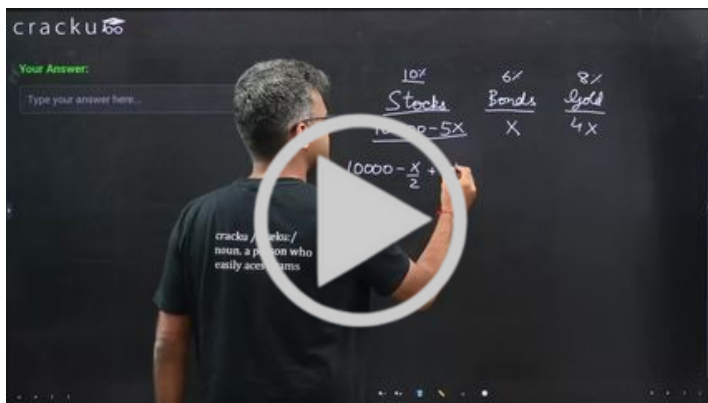
14. Arun, Varun and Tarun, if working alone, can complete a task in 24, 21, and 15 days, respectively. They charge Rs 2160, Rs 2400, and Rs 2160 per day, respectively, even if they are employed for a partial day. On any given day, any of the workers may or may not be employed to work. If the task needs to be completed in 10 days or less, then the minimum possible amount, in rupees, required to be paid for the entire task is

- A 38400
- B 38880
- C 34400
- D 47040



VIDEO SOLUTION

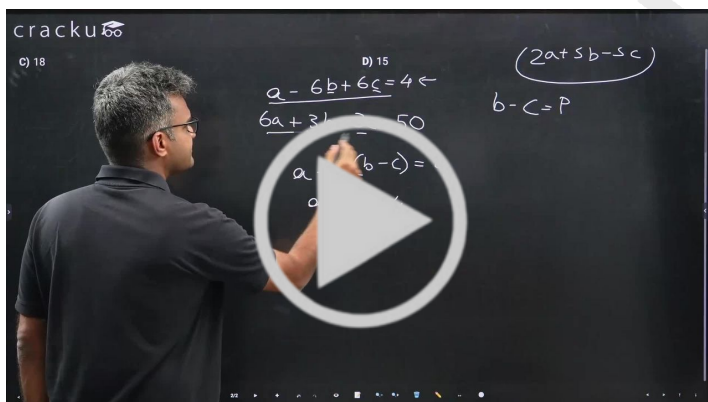
15. Kamala divided her investment of Rs 100000 between stocks, bonds, and gold. Her investment in bonds was 25% of her investment in gold. With annual returns of 10%, 6%, 8% on stocks, bonds, and gold, respectively, she gained a total amount of Rs 8200 in one year. The amount, in rupees, that she gained from the bonds, was



VIDEO SOLUTION

16. If $a - 6b + 6c = 4$ and $6a + 3b - 3c = 50$, where a , b and c are real numbers, the value of $2a + 3b - 3c$ is

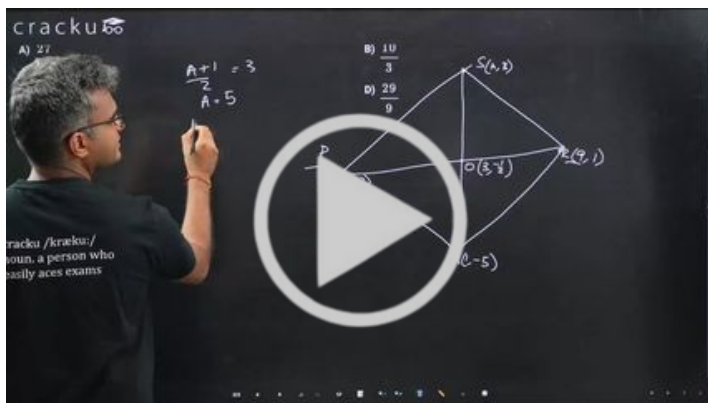
- A 20
- B 14
- C 18
- D 15



VIDEO SOLUTION

17. The (x, y) coordinates of vertices P , Q and R of a parallelogram $PQRS$ are $(-3, -2)$, $(1, -5)$ and $(9, 1)$, respectively. If the diagonal SQ intersects the x -axis at $(a, 0)$, then the value of a is

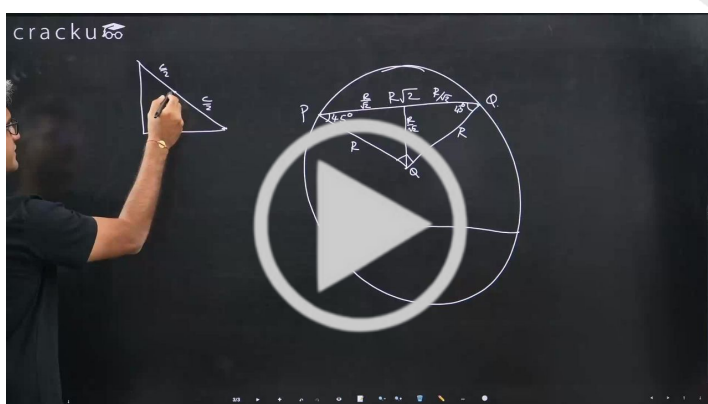
- A $\frac{27}{7}$
- B $\frac{10}{3}$
- C $\frac{13}{4}$
- D $\frac{29}{9}$



VIDEO SOLUTION

18. In a circle with center C and radius $6\sqrt{2}$ cm, PQ and SR are two parallel chords separated by one of the diameters. If $\angle PQC = 45^\circ$, and the ratio of the perpendicular distance of PQ and SR from C is $3 : 2$, then the area, in sq. cm, of the quadrilateral $PQRS$ is

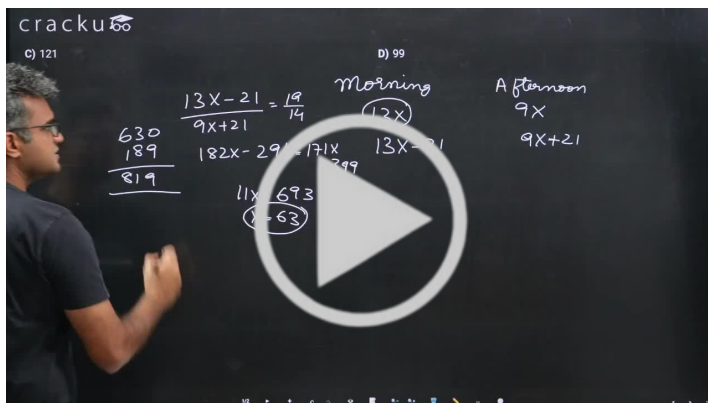
- A $4(3 + \sqrt{14})$
- B $4(3\sqrt{2} + \sqrt{7})$
- C $20(3 + \sqrt{14})$
- D $20(3\sqrt{2} + \sqrt{7})$



VIDEO SOLUTION

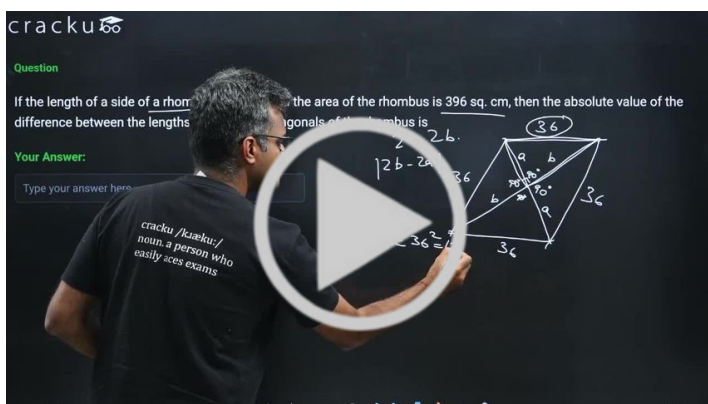
19. The ratio of the number of students in the morning shift and afternoon shift of a school was $13 : 9$. After 21 students moved from the morning shift to the afternoon shift, this ratio became $19 : 14$. Next, some new students joined the morning and afternoon shifts in the ratio $3 : 8$ and then the ratio of the number of students in the morning shift and the afternoon shift became $5 : 4$. The number of new students who joined is

- A 110
- B 88
- C 121
- D 99



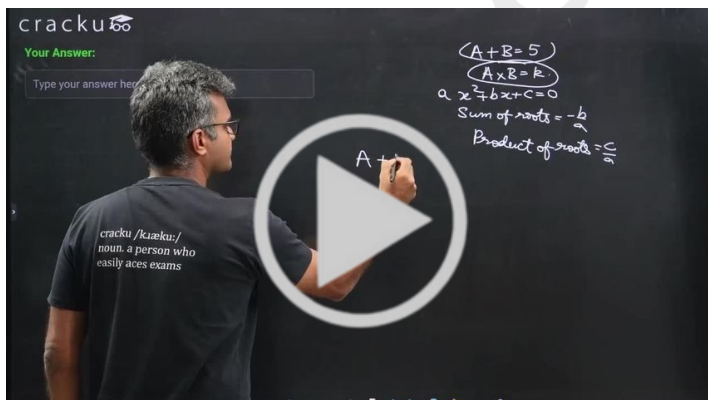
VIDEO SOLUTION

20. If the length of a side of a rhombus is 36 cm and the area of the rhombus is 396 sq. cm, then the absolute value of the difference between the lengths, in cm, of the diagonals of the rhombus is



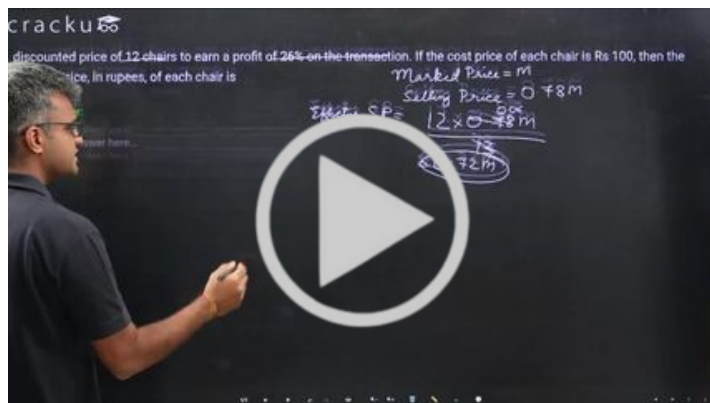
VIDEO SOLUTION

21. The number of non-negative integer values of k for which the quadratic equation $x^2 - 5x + k = 0$ has only integer roots, is



VIDEO SOLUTION

22. A shopkeeper offers a discount of 22% on the marked price of each chair, and gives 13 chairs to a customer for the discounted price of 12 chairs to earn a profit of 26% on the transaction. If the cost price of each chair is Rs 100, then the marked price, in rupees, of each chair is



 VIDEO SOLUTION



Answers

Quant

1.B	2.D	3.6	4.A	5.15	6.A	7.D	8.16
9.A	10.D	11.55	12.A	13.A	14.A	15.900	16.C
17.D	18.C	19.D	20.60	21.3	22.175		

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Explanations

Quant

1. **B**

First function $f(x) = x^2 - 4cx + 8$

For this function $a > 0$, so minimum value will occur at $x = -\frac{b}{2a} = -\left(-\frac{4c}{2}\right) = 2c$

So, the minimum value of the function is $= 2c^2 - 4c(2c) + 8c = -4c^2 + 8c$

Second function $g(x) = -x^2 + 3cx - 2c$

For this function $a < 0$, so maximum value will occur at $x = -\frac{b}{2a} = -\frac{(-3c)}{2} = \frac{3c}{2}$

So, the maximum value of the function is $= -\left(\frac{3c}{2}\right)^2 + 3c\left(\frac{3c}{2}\right) - 2c = \frac{9c^2}{4} - 2c$

So, as per the given condition,

$$\frac{9c^2}{4} - 2c < -4c^2 + 8c$$

$$\text{or, } \frac{9c^2}{4} + 4c^2 < 8c + 2c$$

$$\text{or, } \frac{25c^2}{4} < 10c$$

$$\text{or, } \frac{5c^2}{4} < 2c$$

$$\text{or, } 5c^2 < 8c$$

$$\text{or, } 5c^2 - 8c < 0$$

$$\text{or, } c\left(c - \frac{8}{5}\right) < 0$$

$$\text{or, } 0 < c < \frac{8}{5}$$

So, the value of c which lies in this range is $\frac{1}{2}$

 VIDEO SOLUTION

2. **D**

According to question, Shruti travels a distance of 224 km in four parts for a total travel time of 3 hours.

Given, in the first part time required is 30 minutes.

Also, the time taken to cover these four parts follow arithmetic progression.

So, let us say the times be 30 minutes, $(30+d)$ minutes, $(30+2d)$ minutes and $(30+3d)$ minutes

$$\text{So, } 30 + (30 + d) + (30 + 2d) + (30 + 3d) = 180$$

$$\text{or, } 6d = 180 - 120$$

$$\text{or, } 6d = 60$$

$$\text{or, } d = \frac{60}{6} = 10 \text{ minutes.}$$

So, the time required in these four parts are 30 minutes, 40 minutes, 50 minutes and 60 minutes respectively.

Now, in the first part, speed is 960 metres per minute.

Speed in the four parts is also in arithmetic progression.

Let's say the speeds be $(960 + x), (960 + 2x), (960 + 3x)$ metres per minute

So, total distance covered = $960 \times 30 + (960 + x) \times 40 + (960 + 2x) \times 50 + (960 + 3x) \times 60$ metres

$$\text{So, } 960 \times 30 + (960 + x) \times 40 + (960 + 2x) \times 50 + (960 + 3x) \times 60 = 224000$$

$$\text{or, } 960(30 + 40 + 50 + 60) + 40x + (2x)(50) + (3x)(60) = 224000$$

$$\text{or, } 172800 + 320x = 224000$$

$$\text{or, } 320x = 224000 - 172800 = 51200$$

$$\text{or, } x = \frac{51200}{320} = 160$$

So, the speed in the fourth part is $960 + 3x = 960 + 3 \times 160 = 1440$ metres per minute

Time in the fourth part is 60 minutes

So, distance covered in the fourth part = $1440 \times 60 = 86400$ metres

So, option D is the correct answer.

 VIDEO SOLUTION

3.6

According to question, in the given 3-digit number N, the digits are non-zero and distinct.

So, the possible digits in 3-digit number = 2, 3, 5, 6, 7, 8

It is also given that only one of the digits is a prime number.

So, the minimum possible value of N = 268

(2 is the smallest prime digit, and the non-prime digits has to be 6 and 8)

$$\text{Now, } 268 = 4 \times 67 = 2^2 \times 67$$

$$\text{So, the number of factors} = (2 + 1)(1 + 1) = 3 \times 2 = 6$$

 VIDEO SOLUTION

4.A

For $n=3,4,5$ and $x \in [n, n+1)$ we have $\lfloor x \rfloor = n$, so the equation

$$\lfloor x^2 \rfloor = \lfloor x \rfloor^2 = n^2$$

$$x^2 \in [n^2, n^2 + 1), \text{ i.e. } x \in [n, \sqrt{n^2 + 1})$$

Thus for $3 \leq x \leq 6$

$$S = [3, \sqrt{10}) \cup [4, \sqrt{17}) \cup [5, \sqrt{26}) \cup 6$$

Option B and C have $\sqrt{10}$ included, which is not part of the original set. And Option D has $\sqrt{18}$. So, it is not possible.

Option A is the answer.

 VIDEO SOLUTION

5.15

Let the number of stocks B hold be x

So, number of stocks C hold is $20 - x$

$$\text{So, } 10 \times 120 + 90 \times x + 150 \times (20 - x) = 3300$$

$$\text{or, } 1200 + 90x + 150(20 - x) = 3300$$

$$\text{or, } 1200 + 90x + 3000 - 150x = 3300$$

$$\text{or, } 4200 - 60x = 3300$$

$$\text{or, } 60x = 900$$

$$\text{or, } x = \frac{900}{60} = 15$$

So, the number of shares of stock that B hold is 15.

 VIDEO SOLUTION

6. A

Given expression is $\{(a_1)^1 \times (a_2)^2 \times \dots \times (a_{20})^{20}\} < \{a_{21} \times a_{22} \times \dots \times a_{20+m}\}$,

$$\{(a_1)^1 \times (a_2)^2 \times \dots \times (a_{20})^{20}\} = \{3^1 \times 3^4 \times 3^9 \dots \times 3^{400}\}$$

Sum of square of n natural numbers is $\frac{n \cdot (n+1) \cdot (2n+1)}{6}$

$$\frac{(20 \cdot 21 \cdot 41)}{6} = 3^{2870}$$

On right hand side of inequality we have $\{a_{21} \times a_{22} \times \dots \times a_{20+m}\}$

$$= 3^{21} \times 3^{22} \times \dots \times 3^{20+m} = 3^{21+22+\dots+20+m}$$

Using the sum of the first (n) natural numbers,

$$1 + 2 + \dots + n = \frac{n(n+1)}{2}$$

$$21 + 22 + \dots + (20 + m)$$

$$= 1 + 2 + \dots + (20 + m) - (1 + 2 + \dots + 20)$$

$$1 + 2 + \dots + (20 + m) = \frac{(20+m)(21+m)}{2}$$

$$1 + 2 + \dots + 20 = \frac{20 \cdot 21}{2} = 210$$

$$\text{So, } 21 + 22 + \dots + (20 + m)$$

$$= \frac{(20+m)(21+m)}{2} - 210$$

$$\text{Expanding, } (20 + m)(21 + m) = m^2 + 41m + 420$$

$$\text{Thus, } \frac{m^2 + 41m + 420}{2} - 210$$

$$= \frac{m^2 + 41m}{2}$$

Since the bases are equal, we must compare the powers.

$$2870 < \frac{m^2 + 41m}{2} \Rightarrow 5740 < m(m + 41)$$

Here, we can put in the option to check the minimum value that satisfies the inequality.

56: We get $5740 < 5264$. This is false

57: We get $5740 < 5586$. This is false

58: We get $5740 < 5742$. This is the minimum possible value.

7.D

For base of log in range $1/4 \in (0, 1)$ and $\log_{1/4}(x) > 0$ is true only if $0 < x < 1$.

For integer n , $x = n^2 - 7n + 11$ is an integer, so it cannot lie strictly between 0 and 1.

So, there is no integer value for which this inequality is satisfied.

8.16

It is given, $x > y \geq 3$ and $x + y < 14$

Now for $y = 3$, the values of x can be 4,5,6,7,8,9,10 (7 cases)

Then for $y = 4$, the values of x can be 5,6,7,8,9 (5 cases)

Then for $y = 5$, the values of x can be 6,7,8 (3 cases)

Then for $y = 6$, the values of x will be 7 (1 case)

So, total number of cases = $1+3+5+7=16$ cases

9.A

Let the principal be ₹ P and rate of interest be $r\%$.

$$\text{Now, } 13920 = P + \frac{P \times r \times 3}{100}$$

$$\text{or, } 13920 - P = \frac{P \times r \times 3}{100} \text{ ---->(1)}$$

$$\text{Also, } 18960 = P + \frac{P \times r \times 13}{100 \times 2}$$

$$18960 - P = \frac{P \times r \times 13}{100 \times 2} \text{ ---->(2)}$$

Dividing eqn(1) by eqn(2),

$$\frac{13920 - P}{18960 - P} = \frac{3}{\frac{13}{2}} = \frac{6}{13}$$

$$\text{or, } (13920 - P) 13 = (18960 - P) 6$$

$$\text{or, } 13920 \times 13 - 13P = 18960 \times 6 - 6P$$

$$\text{or, } 13920 \times 13 - 18960 \times 6 = 13P - 6P$$

$$\text{or, } 180960 - 113760 = 7P$$

$$\text{or, } 67200 = 7P$$

$$\text{or, } P = \frac{67200}{7} = 9600$$

Putting this in equation (1),

$$13920 - 9600 = \frac{9600 \times r \times 3}{100}$$

$$\text{or, } 4320 = 96 \times 3r$$

$$\text{or, } r = \frac{4320}{96 \times 3} = 15$$

So, rate percent is 15%

Now if the same sum had been invested for 2 years at the same rate as before but with interest compounded

$$\text{every 6 months, amount} = 9600 \left(1 + \frac{\frac{15}{2}}{100}\right)^4 = 9600 \left(1 + \frac{7.5}{100}\right)^4$$

$$\text{So, interest} = 9600 \left(1 + \frac{7.5}{100}\right)^4 - 9600$$

$$= \text{Rs } 3220.50$$

$$= \text{Rs } 3221$$

So, the total interest earned is Rs 3221.

 VIDEO SOLUTION

10. D

$$200 \text{ L with } (30\%) \text{ acid} \Rightarrow \text{acid} = 0.30 \times 200 = 60L$$

$$\text{Replace 20\% with water: remaining acid} = 60 \times (1 - 0.20) = 60 \times 0.8 = 48L.$$

$$\text{Replace 10\% with pure acid: After removing 10\% of the mixture, the acid becomes } 48 \times 0.9 = 43.2L, \text{ then adding back 20 L pure acid} \Rightarrow \text{acid} = 43.2 + 20 = 63.2L.$$

$$\text{Replace 15 with water: acid left} = 63.2 \times (1 - 0.15) = 63.2 \times 0.85 = 53.72L.$$

$$\text{Final concentration} = \frac{53.72}{200} = 0.2686 \approx 26.86.$$

The answer is 27%

 VIDEO SOLUTION

11. 55

Let say number of girls be g and number of boys be b .

$$\text{If 40\% of the girls left, remaining number of girls} = 0.6g$$

$$\text{Also if 60\% of the boys left, remaining number of boys} = 0.4b$$

$$\text{or, } 0.6g = 0.4b + 8$$

$$\text{or, } 6g = 4b + 80$$

$$\text{or, } 3g = 2b + 40$$

$$\text{So, the possible values of } (b, g) \text{ are: } (13, 22), (16, 24), (19, 26), (22, 28), (25, 30), \dots$$

Now, $0.6g$ and $0.4b$ has to be an integer.

So, for this g and b has to be a multiple of 5

$$\text{So, } b = 25 \text{ and } g = 30$$

$$\text{So, minimum possible number of students} = 25 + 30 = 55$$

 VIDEO SOLUTION

12. A

$$\text{Number of ways to choose a sandwich} = {}^5C_1 \text{ ways}$$

$$\text{Number of ways to choose a bread} = {}^4C_1 \text{ ways}$$

Number of ways to choose bread size = 2C_1 ways

Number of ways to choose sauces = ${}^6C_0 + {}^6C_1 + {}^6C_2 = 1 + 6 + 15 = 22$ ways

So, number of different ways = ${}^5C_1 \times {}^4C_1 \times {}^2C_1 \times 22 = 5 \times 4 \times 2 \times 22 = 880$ ways.

 VIDEO SOLUTION

13. **A**

The sum of all the elements in the given set = Sum of first 29 odd numbers = $29^2 = 841$

Let's assume that k is the m_{th} term. Sum of terms less than k = sum of first (m-1) odd numbers = $(m-1)^2$

$$841 - m^2 = (m-1)^2$$

$$841 - m^2 = m^2 - 2m + 1$$

$$840 - 2m^2 + 2m = 0$$

$$m^2 - m - 420 = 0$$

$$(m-21)(m+20) = 0$$

$$m = 21 - 20$$

$$m = 20. \text{ And the 20th term is } 2 \times m + 1 = 41$$

 VIDEO SOLUTION

14. **A**

Let's assume that Total work = 1

If the work is entirely carried by each one of them it would cost

$$\text{Arun: } 2160 \times 24 = 51840$$

$$\text{Varun: } 2400 \times 21 = 50400$$

$$\text{Tarun: } 2160 \times 15 = 32400$$

The cheapest worker is Tarun. He can finish the work in 15 days. But we need the work completed in 10 days.

$$\text{Tarun does } 10 \times \frac{1}{15} = \frac{2}{3}$$

$$1 - \frac{2}{3} = \frac{1}{3}$$

Complete the remaining (1/3) work using the next cheapest worker, which is Varun.

$$\text{Days needed: } \frac{x}{21} = \frac{1}{3} \Rightarrow x = 7$$

$$\text{Total cost} = \text{Amount paid for Tarun} + \text{Amount paid for Varun} = 10 \times 2160 + 7 \times 2400 = 38400$$

 VIDEO SOLUTION

15. **900**

Let the amounts invested in Stocks be S, Bonds B, and, Gold be G

$$\text{Given that } S + B + G = 100000, B = 0.25G$$

$$0.10S + 0.06B + 0.08G = 8200$$

$$\text{Substitute } S = 100000 - B - G = 100000 - 1.25G$$

$$0.10(100000 - 1.25G) + 0.06(0.25G) + 0.08G = 8200$$

$$10000 - 0.125G + 0.015G + 0.08G = 8200$$

$$10000 - 0.03G = 8200$$

$$0.03G = 1800 \Rightarrow G = 60000$$

$$B = 0.25G = 15000$$

$$\text{Gain from bonds} = 0.06 \times 15000 = 900$$

 VIDEO SOLUTION

16. **C**

$$\text{Given, } a - 6b + 6c = 4 \text{ --->(1)}$$

$$\text{and, } 6a + 3b - 3c = 50 \text{ --->(2)}$$

$$\text{Multiplying eqn (1) with } x, ax - 6bx + 6cx = 4x$$

$$\text{And multiplying eqn (2) with } y, 6ay + 3by - 3cy = 50y$$

$$\text{So, } x + 6y = 2 \text{ and } 3y - 6x = 3$$

$$\text{So, } x + 6y = 2 \text{ --->(3) and } -2x + y = 1 \text{ --->(4)}$$

$$\text{Multiplying eqn (4) with 6 and subtracting from eqn (3),}$$

$$13x = -4$$

$$\text{So, } x = -\frac{4}{13}$$

$$\text{Putting the value of } x \text{ in equation (4),}$$

$$y = \frac{5}{13}$$

$$\text{So, the final answer is } 4x + 5y$$

$$= 4\left(-\frac{4}{13}\right) + 5\left(\frac{5}{13}\right) = -\frac{16}{13} + \frac{25}{13} = \frac{9}{13} = 18$$

$$\text{So, correct answer is 18.}$$

 VIDEO SOLUTION

17. **D**

$$\text{Given } (P(-3,-2)), (Q(1,-5)), \text{ and } (R(9,1)).$$

$$\text{For parallelogram (PQRS), } S = P + R - Q = (-3,-2) + (9,1) - (1,-5) = (5,4)$$

$$\text{Diagonal SQ passes through } S(5,4) \text{ and } Q(1,-5).$$

$$\text{Slope } m = \frac{-5 - 4}{1 - 5} = \frac{9}{4}$$

$$\text{Equation of SQ is } y - 4 = \frac{9}{4}(x - 5)$$

$$\text{At the x-axis, } y = 0: -4 = \frac{9}{4}(a - 5)$$

$$-16 = 9(a - 5)$$

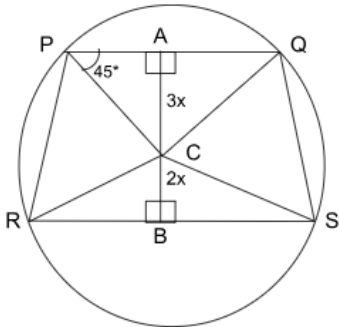
$$9a = 29$$

$$a = \frac{29}{9}$$

$$\text{So, } a = \frac{29}{9}$$

18.C

The radius of the circle is given to be $6\sqrt{2}$ cm. We are also given that the ratio of CA to CB is 3:2. So, if we assume the value of CA to be $3x$, then the value of CB becomes $2x$. We are also given that $\angle PQC = 45^\circ$ and since PCQ is an isosceles triangle, $\angle CPQ = 45^\circ$ and $\angle PCQ = 90^\circ$.



So, the triangle PCQ is right-angled at C. The value of PQ can be calculated using Pythagoras' theorem as,

$$PQ^2 = PC^2 + QC^2$$

$$PQ^2 = (6\sqrt{2})^2 + (6\sqrt{2})^2 = 72 + 72 = 144$$

$$PQ = 12 \text{ cm}$$

$$AQ = \frac{PQ}{2} = \frac{12}{2} = 6 \text{ cm}$$

In triangle ACQ,

$$AC^2 + AQ^2 = CQ^2$$

$$(3x)^2 + (6)^2 = (6\sqrt{2})^2$$

$$9x^2 + 36 = 72$$

$$9x^2 = 36$$

$$x^2 = 4$$

$$x = 2$$

$$CB = 2x = 2 \times 2 = 4 \text{ cm}$$

In triangle BCS,

$$BC^2 + BS^2 = CS^2$$

$$(4)^2 + (BS)^2 = (6\sqrt{2})^2$$

$$16 + (BS)^2 = 72$$

$$BS^2 = 56$$

$$BS = 2\sqrt{14} \text{ cm}$$

$$RS = 2 \times BS = 2 \times 2\sqrt{14} = 4\sqrt{14} \text{ cm}$$

The area of the quadrilateral PQRS can be calculated as,

$$\text{Area} = \frac{1}{2} \times AB \times (PQ + RS) = \frac{1}{2} \times 10 (12 + 4\sqrt{14}) = 20 (3 + \sqrt{14}) \text{ cm}^2$$

Hence, the correct answer is option C.

VIDEO SOLUTION

19. D

Let $M=13k$, $A=9k$.

After 21 students were moved $\frac{13k-21}{9k+21} = \frac{19}{14} \Rightarrow 14(13k - 21) = 19(9k + 21)$

So $182k - 294 = 171k + 399 \Rightarrow 11k = 693 \Rightarrow k = 63$

So, the number of students in the morning and afternoon shifts are $819 - 21 = 798$, $567 + 21 = 588$ respectively.

Let's assume that $3t$ and $8t$ students joined the respective sessions.

$$\frac{798 + 3t}{588 + 8t} = \frac{5}{4} \Rightarrow 4(798 + 3t) = 5(588 + 8t)$$

So $3192 + 12t = 2940 + 40t \Rightarrow 28t = 252 \Rightarrow t = 9$

Number of new students = $11t = 99$

VIDEO SOLUTION

20. 60

Let the diagonals of the rhombus be d_1 and d_2

We know that the Area = $\frac{1}{2}d_1d_2 = 396 \Rightarrow d_1d_2 = 792$

For a rhombus with side $a=36$, the diagonals intersect at right angle. Giving a right-angle triangle with the side as hypotenuse.

$$\left(\frac{d_1}{2}\right)^2 + \left(\frac{d_2}{2}\right)^2 = a^2 \Rightarrow \frac{d_1^2 + d_2^2}{4} = 36^2 = 1296$$

$$d_1^2 + d_2^2 = 5184$$

We want $|d_1 - d_2|$

$$(d_1 - d_2)^2 = d_1^2 + d_2^2 - 2d_1d_2 = 5184 - 2 \cdot 792 = 5184 - 1584 = 3600$$

$$|d_1 - d_2| = \sqrt{3600} = 60$$

VIDEO SOLUTION

21. 3

The given quadratic equation is $x^2 - 5x + k = 0$

Now, discriminant $D = 5^2 - 4k = 25 - 4k$

Now, it is given the equation must have integer roots.

So, $25 - 4k$ has to be a perfect square.

We need to find non-negative integer values of k

Now, for $k = 0$, $D = 25 - 4 \times 0 = 25$, is a perfect square

For $k = 4$, $D = 25 - 4 \times 4 = 25 - 16 = 9$, is a perfect square

For $k = 6$, $D = 25 - 4 \times 6 = 1$, is a perfect square

So, there are three non negative integer values of k .

So, correct answer is 3.

 VIDEO SOLUTION

22.175

Cost price of each chair = 100

For 13 chairs, total cost = $13 \times 100 = 1300$

Profit = 26%, so total revenue

$$1.26 \times 1300 = 1638$$

We were told that this amount is equal to the discounted price of 12 chairs. So the discounted selling price per chair = $SP_{\text{disc}} = \frac{1638}{12} = 136.5$

Discount offered = 22%, so:

$$MP = \frac{136.5}{0.78} = 175$$

 VIDEO SOLUTION

